

ADVANCED SOFTWARE ENGINEERING

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Project Plan

Outline

- ▣ Software Project Management
- ▣ The Planning Game
- ▣ Product Vision
- ▣ Product Roadmap
- ▣ Release Plan
- ▣ Iteration Plan
- ▣ Daily Standup

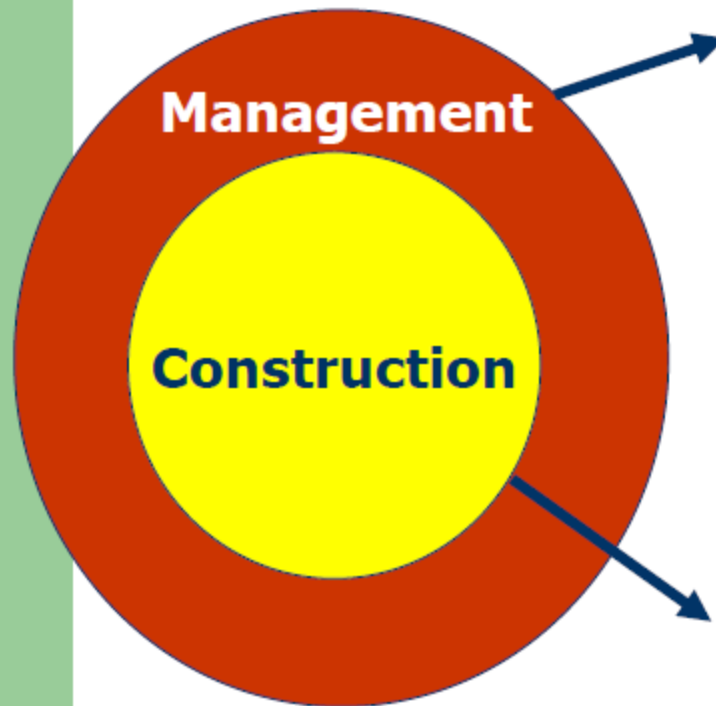
Software Project Management

- ▣ *Software Project Management* concerned with activities involved in ensuring that software is delivered on time and on schedule and in accordance with the requirements of the organisations developing and obtaining the software.

Software Project Plan

- ▣ *Software Project Plan* is an aspect of Project Management, which comprises of various processes. The aim of these processes is to ensure that various Project tasks are well coordinated and they meet the various project objectives including timely completion of the project.

Software Development



**project planning and
management
configuration
management
quality assurance
installation and training
etc.**

**Requirements
Design
Coding
Testing
Maintenance
etc.**

Project Planning

In general software project planning answers the following question :

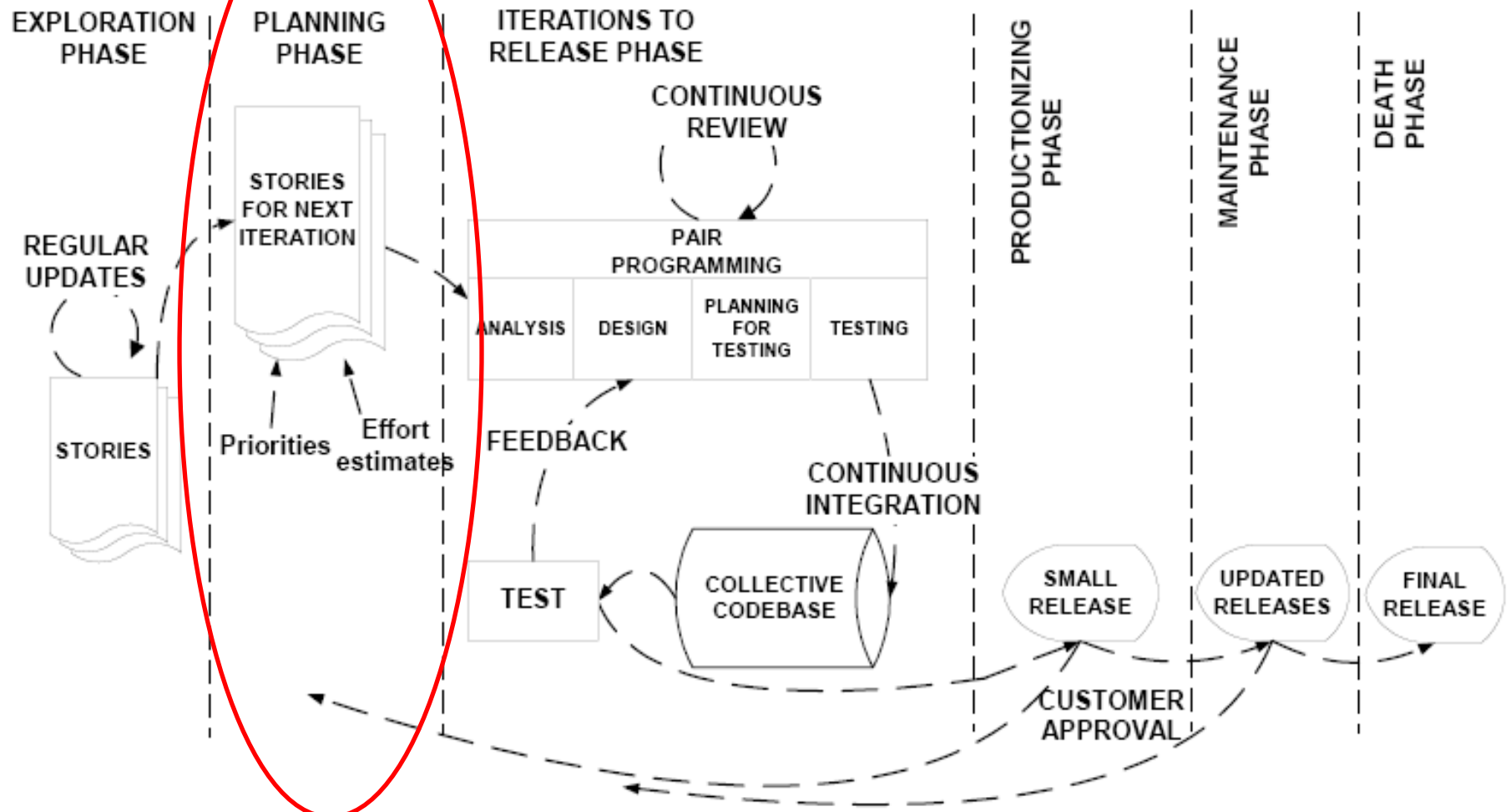
- ▣ How much will it cost?
- ▣ How long will it take?
- ▣ How many people will it take?
- ▣ What kind of people will it take?
- ▣ What might go wrong?
- ▣ What can we do to minimize chance of failure?

Project Planning

It includes the following tasks:

- ▣ **Scoping** - understand problem and the work that must be done
- ▣ **Estimation** - time and effort
- ▣ **Risk** - what can go wrong? What can we do about it?
- ▣ **Schedule** - resources and milestones
- ▣ **Control strategy** - how can we tell when things are going wrong?

Planning in the XP lifecycle



The Planning Game

Create a little emotional distance from planning by treating it as a game (hence the name). The game has a goal, playing pieces, players, and rules for allowable moves.

: The goal of the game is to put the greatest possible value of stories into production over the life of the game.

: The players are Business and Development

The Planning Game

- ▣ : The basic playing piece is the UserStory. Stories have a value and a cost.
- ▣ : There are three phases to the game.
 - —Find out what new things the system could do. Ex: *Write a story, Estimate a story, Split a story.*
 - —Decide what subset of all possible requirements to pursue next. Ex: *Sort stories by value, Sort stories by risk, Set velocity, Choose set of cards in the release.*
 - —Guide development as reality molds the plan. Ex: *Iteration plan, reestimate all of the remaining stories and set velocity again.*

The Planning Game

- ▣ Every XP team includes several business experts — the *on-site customers* — *who are responsible for making business decisions.*
- ▣ **Good agile teams plan.** They just don't plan more than they need to (or less). They wait until the last responsible moment to make the commitment / decision.

The Planning Game

need to decide about

- ▣ **Scope**—How much of a problem must be solved for the system to be valuable in production? The business person is in a position to understand how much is not enough and how much is too much.
- ▣ **Priority**—If you could only have A or B at first, which one do you want? The business person is in a position to determine this, much more so than a programmer.
- ▣ **Composition of releases**—How much or how little needs to be done before the business is better off with the software than without it? The programmer's intuition about this question can be wildly wrong.
- ▣ **Dates of releases**—What are important dates at which the presence of the software (or some of the software) would make a big difference?

The Planning Game

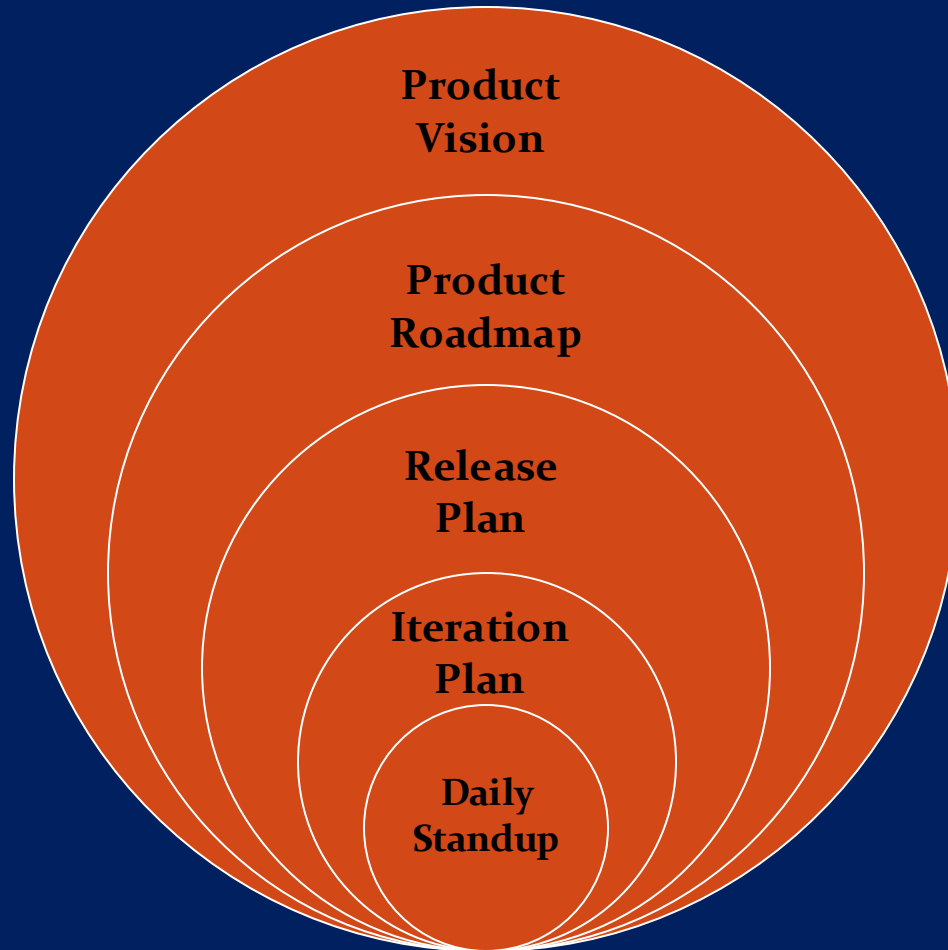
Business can't make these decisions in a vacuum. Development needs to make the technical decisions that provide the raw material for the business decisions. Technical people decide about

- ▣ **Estimates**—How long will a feature take to implement?
- ▣ **Consequences**—There are strategic business decisions that should be made only when informed about the technical consequences. Choice of a database is a good example. Development needs to explain the consequences.

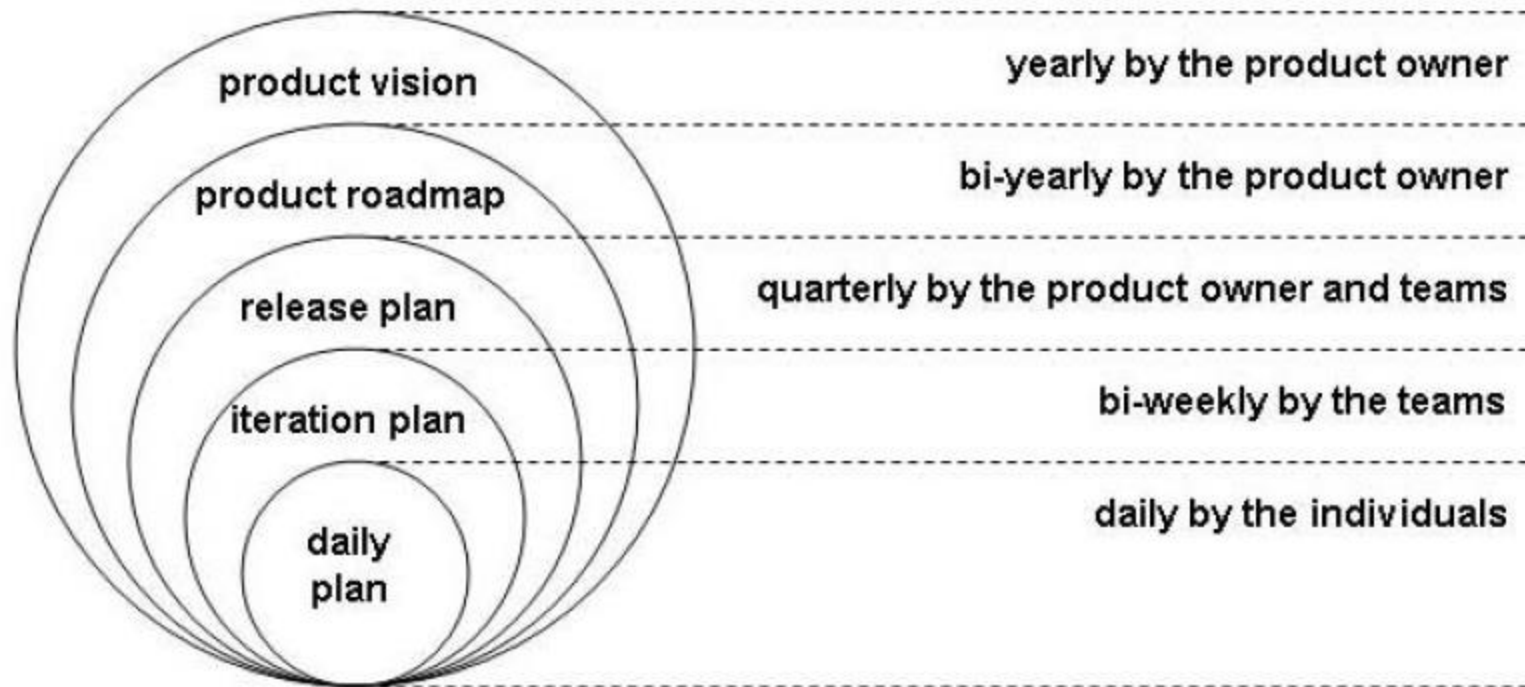
The Planning Game

- ▣ **Process**—How will the work and the team be organized?
- ▣ **Detailed scheduling**—Within a release, which stories will be done first? The programmers need the freedom to schedule the riskiest segments of development first, to reduce the overall risk of the project. Within that constraint, they still tend to move business priorities earlier in the process, reducing the chance that important stories will have to be dropped toward the end of the development of a release.

5 Levels of Planning

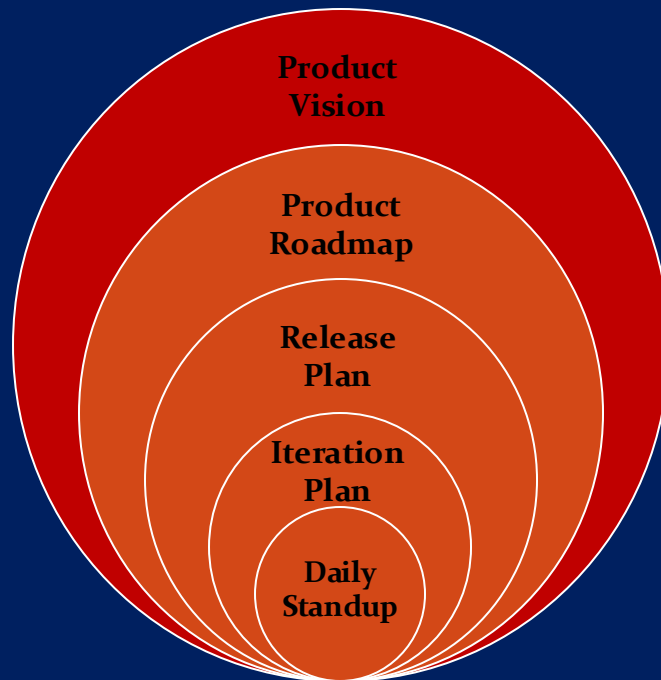


5 Levels of Planning



Product Vision

- ▣ A vision statement is a high-level statement that describes the goal of the system.



Product Vision

- ▣ Without this level, there's a really good chance that your good work won't amount to business advantage. Vision is a starting point. It needs to adjust to what you learn by interacting with the market.

Product Vision

- ▣ In this vision the product owner explains
Who's the **target customer**? What
problem are you solving? Who is the
competition? What are your
strengths/weaknesses?

Product Vision: Example

*“ For a mid-sized company's marketing and sales departments **who** need basic CRM (Customer Relation Management) functionality, **the CRM-Innovator** is a Web-based service **that** provides sales tracking, lead generation, and sales representative support features that improve customer relationships at critical touch points. **Unlike** other services or package software products, **our** product provides very capable services at a moderate cost”*

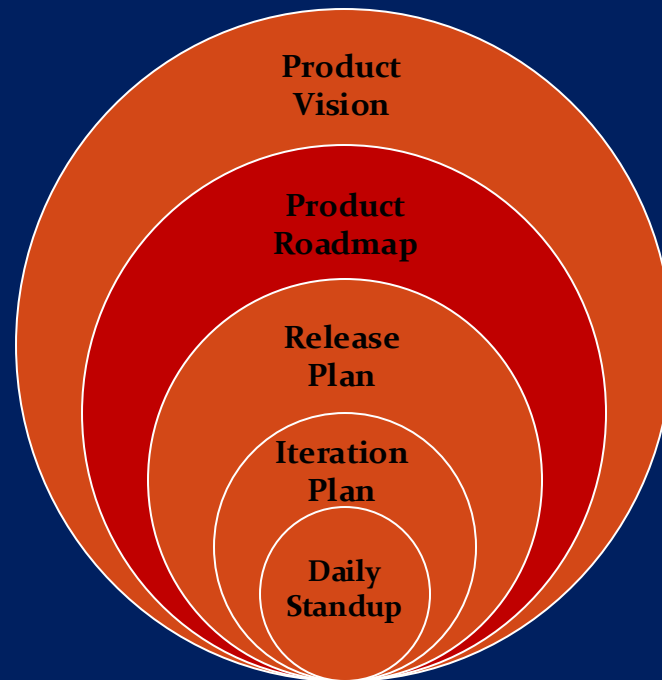
Product Vision- Example

- ▣ The previous example follows the form (which is called “**elevator statement**”):
 - For (target customer)
 - Who (statement of the need or opportunity)
 - The (product name) is a (product category)
 - That (key benefit, compelling reason to buy)
 - Unlike (primary competitive alternative)
 - Our product (statement of primary differentiation)

Another example

- ▣ For dentists and their assistants
who need to efficiently schedule appointments
Dental Clinic 2.0 is desktop and web-based
appointment scheduling software
that supports office and remote access.
Unlike competitive products,
Dental Clinic 2.0 is easy to use and aggressively
priced.

Product Roadmap



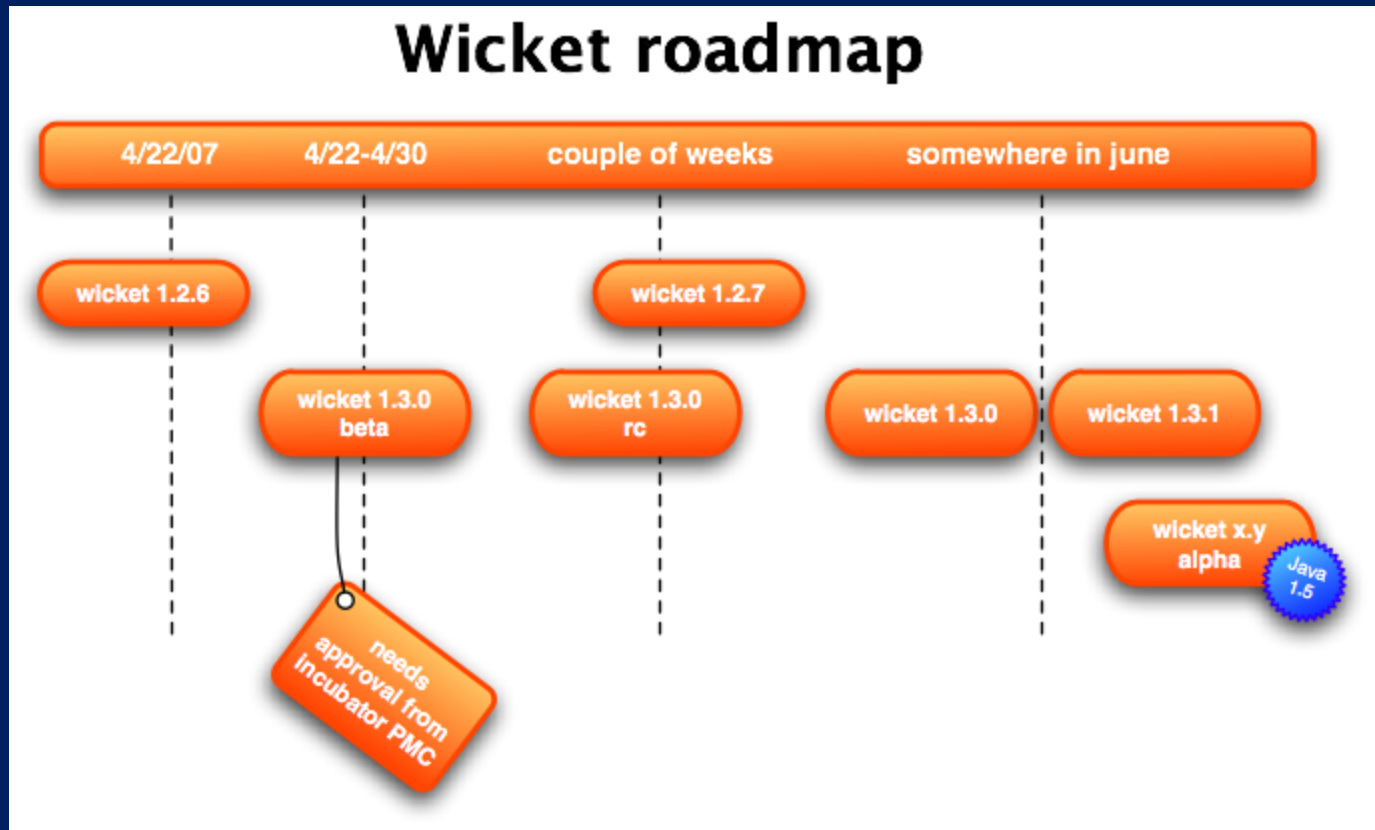
Product Roadmap

- ▣ A roadmap is a planned future, laid out in broad hits — i.e. planned or proposed product releases, listing high level functionality or release themes, for a period usually extending for 2 or 3 significant feature releases into the future.
- ▣ It gives your customers a feel for what you're about and where you're going.

Product Roadmap

- ▣ For startups or companies in fast moving and growing markets, those 2-3 releases might only cover the next 12 months of time. For other more mature companies in less dynamic markets, those releases might cover several years.

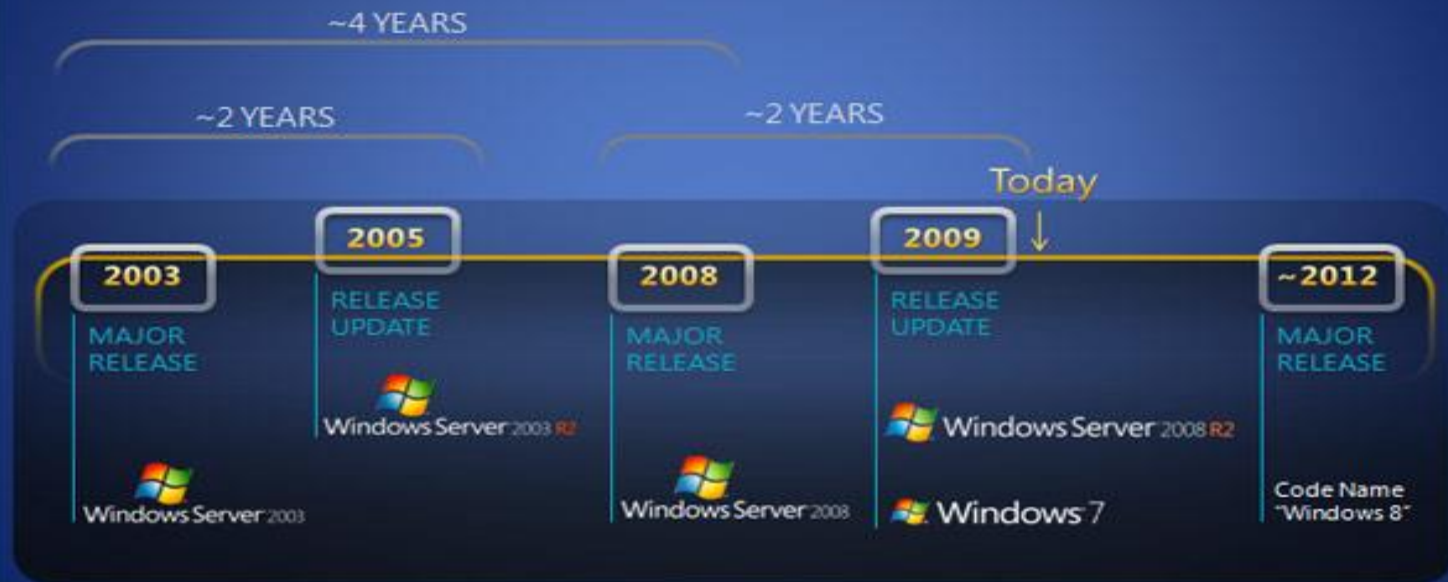
Product Roadmap- Example



Product Roadmap- Example

Windows Server Release Cadence

Delivering on the Promise of New Value



Product Roadmap- Example

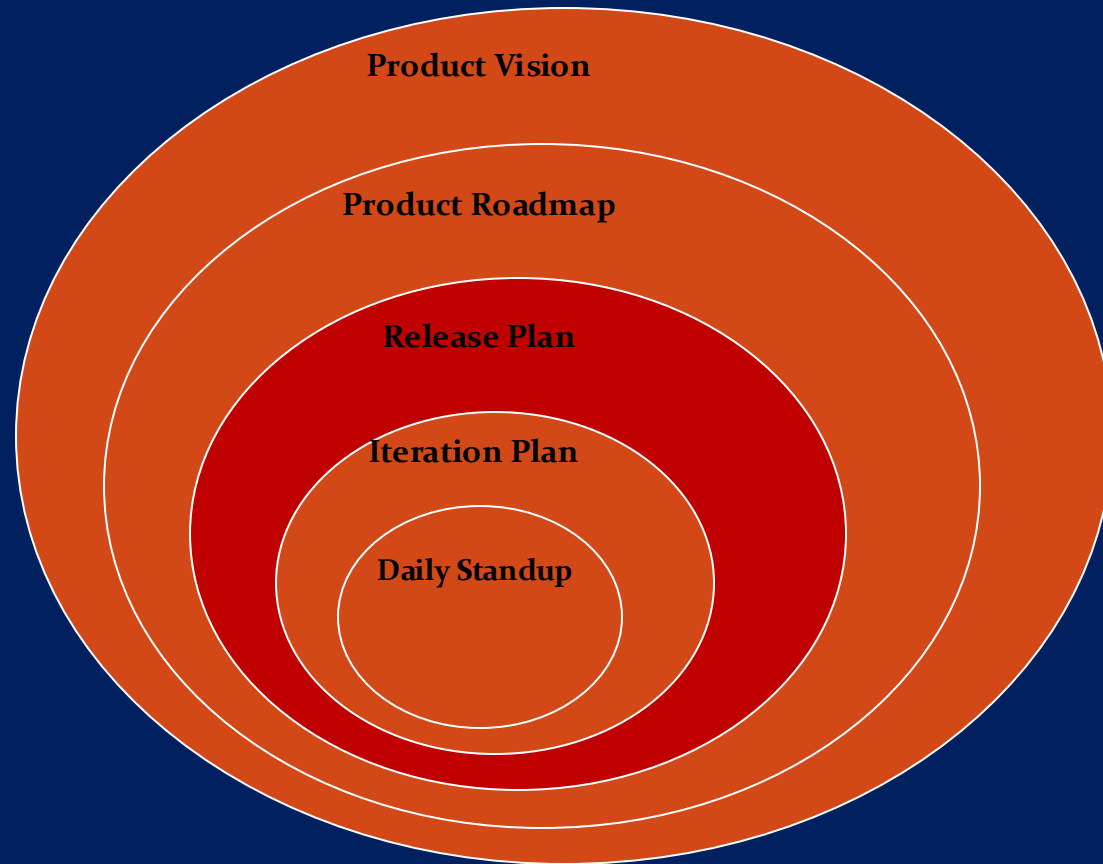


Product Roadmap

The goals for product roadmap are for the product owner to:

- ▣ Communicate the whole
- ▣ Determine and communicate when releases are needed
- ▣ Determine what functionality is sufficient for each release
- ▣ Focus on business value derived from the releases

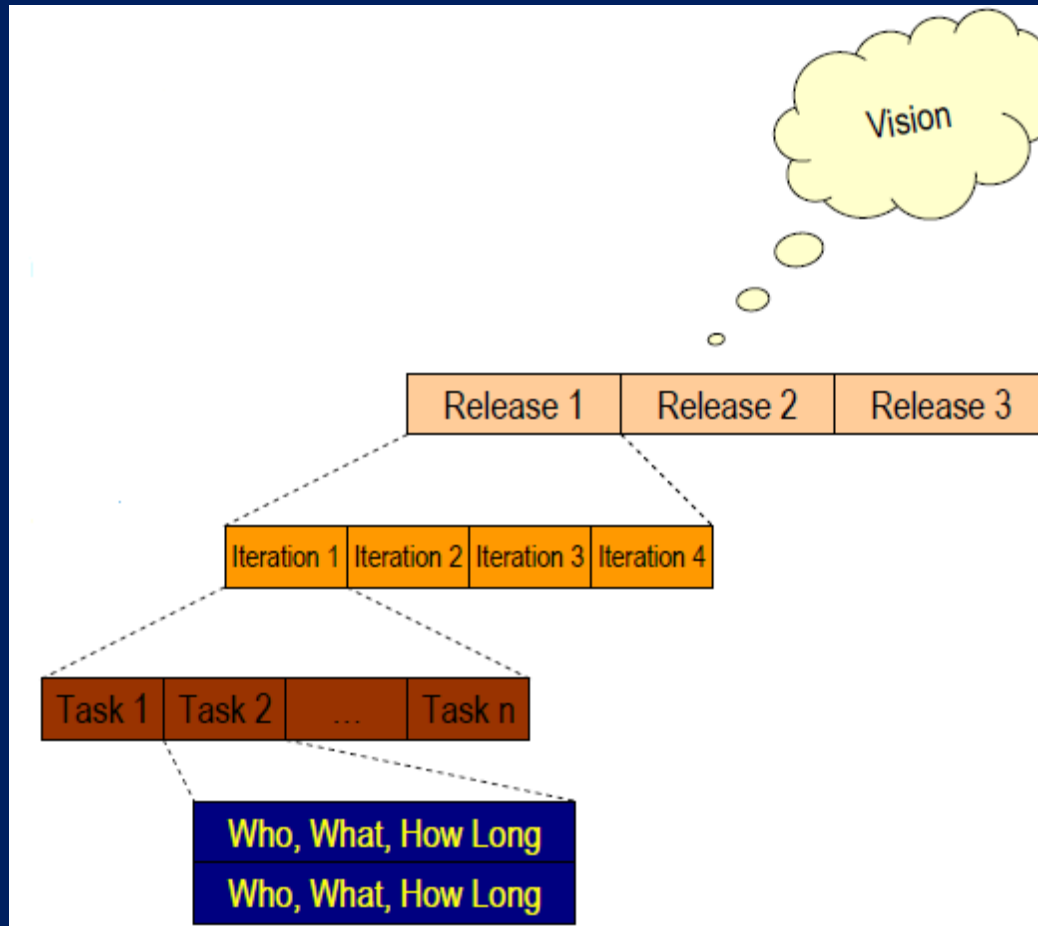
Release Plan



Release Plan

- ▣ **Release Plan:** Projects are divided into Releases which define the project versions that will be published.
- ▣ A **release** is a version of a system with enough new features that it can be delivered to users outside the development group. A release represents perhaps one to three months of work.
- ▣ A project has as many releases as are deemed necessary by the customer and developers. Typically, Releases consist of 80 (+-20) User Stories, broken down into a number of Iterations.
- ▣ **The result is the release plan:** a list of stories (requirements) prioritized by business value.

XP Project



XP Project

- ▣ **Vision** a high-level statement .
- ▣ Within a **release**, XP uses one- to four-weeks iterations of customer requested features for fine-grained feedback about progress.
- ▣ Within an **iteration**, with one- to three-day tasks, so the team can solve problems even during an iteration.
- ▣ Iteration are further broken down into programming **Tasks**, each of 1 to 3 ideal days duration.

Release Plan

- ▣ You may plan by **time** or by **scope**. The project velocity is used to determine either how many stories can be implemented before a given date (time) or
- ▣ How long a set of stories will take to finish (scope).

Release Plan

- ▣ The customer creates the release plan as a list of features in priority order, grouped into iterations according to the estimate and the velocity.
- ▣ For example, assuming a declared velocity of 40 hours per iteration and a team planning to do 3 iterations:

Release Plan - Example

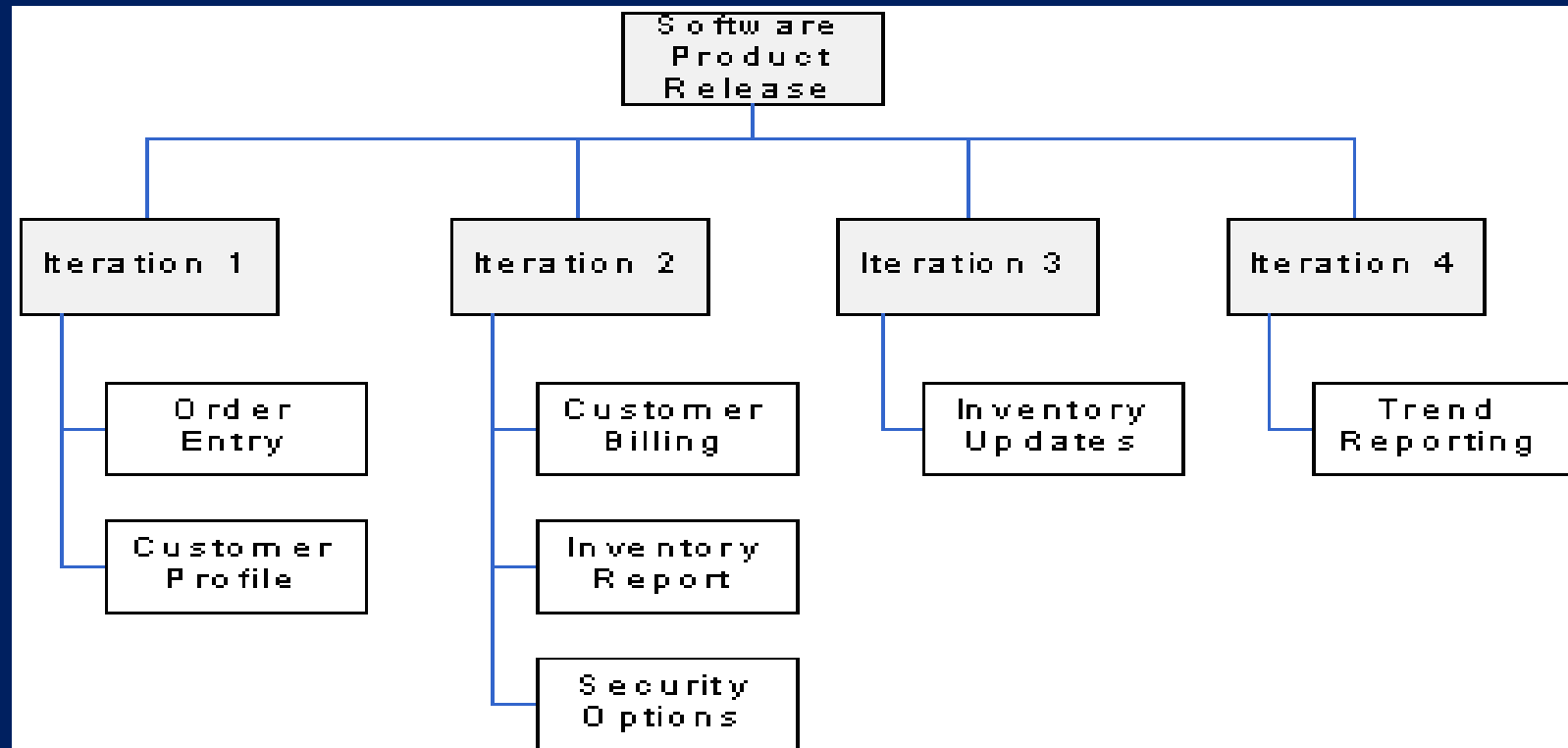
Iteration	Stories	Estimated time
Iteration 1	Text screen	10 hours
	Simple query	30 hours
Iteration 2	Boolean query	20 hours
	Simple screen	20 hours
Iteration 3	Printing	30 hours
	Distribution	10 hours

Release Plan

- ▣ *A scope plan defines the features the team will build in advance, but the release date is uncertain.*

Story	Time Estimate (Ideal Weeks)	Assigned Iteration #	Assigned Release #
Find lowest fare.	3	2	1
Show available flights.	2	1	1
Sort available flights by convenience.	4		2
Purchase ticket.	2	1	1
Do customer profile.	4		
Do simple customer profile.	2	1	1
Do full customer profile.	3		
Review itineraries.	1	2	1
Cancel itinerary.	2		2
Print immigration paperwork.	4		
Show hotels.	3		2
Show hotel availability.	2 ^[a]		2
Offer sophisticated hotel search.	3		
Book a hotel.	1	2	1
Provide hotel/spaceline programs.	3		
Offer airplane hire.	3		

Another example of release plan



Velocity

- ▣ **Calendar Time**: Overall amount of time that passes on clock or calendar
- ▣ **Ideal time** : amount of time something takes when stripped of all interpretation such as meetings, phone calls, emails, trainings , reviews, bug fix, support, sick time, etc.
- ▣ **Velocity** is a simple way of mapping estimates to the calendar. It's the total of the estimates for the stories finished in an iteration.

Release planning

- ▣ Release planning is a joint effort between the customer and the programmers. The customer drives release planning, and the programmers help navigate. As follows:

Release planning

The customer

- ▣ Defines the user stories
- ▣ Decides what business value the stories have
- ▣ Decides what stories to build in this release

The programmers

- ▣ Estimate how long it will take to build each story
- ▣ Warn the customer about significant technical risks
- ▣ Measure their team progress to provide the customer with an overall budget

Release planning

Business Value

- ▣ So we need different factors for making our choice. The first of these is business value. Do those stories first that yield the most benefit to the customers.
- ▣ We want to get a release to the customer as soon as possible. We want this release to be as valuable to the customer as possible. So we give him the things he wants most. That way we can release quickly and the customer feels the benefit.

Release planning

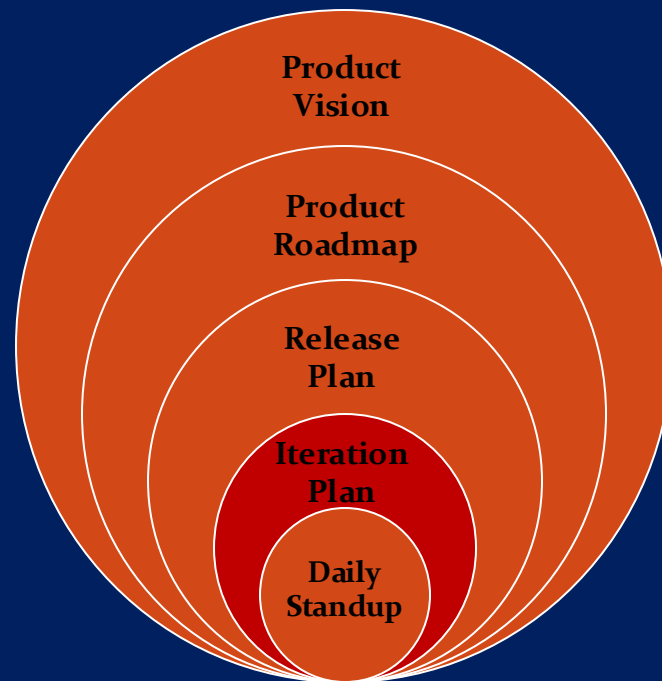
- ▣ How do we measure the business value of a story?
- ▣ The short answer is that we don't; that is, the developers and project managers don't.
- ▣ Determining business value is a decision entirely within the realm of the business people. It's their business; they know what the value is. At least, they are better positioned to guess than we are.
- ▣ The developers can suggest alternatives and ask questions. But in the end business value is always a business call.

Ordering the stories

Prioritizing User Stories

- ▣ It doesn't really help to ask for absolute rankings of stories such as high, medium, and low. All the stories end up being high or the customer wouldn't have written them, now would they? Instead, the customer needs to prepare to answer the question "What would you like us to implement first, and what will we implement later? " based on business value.

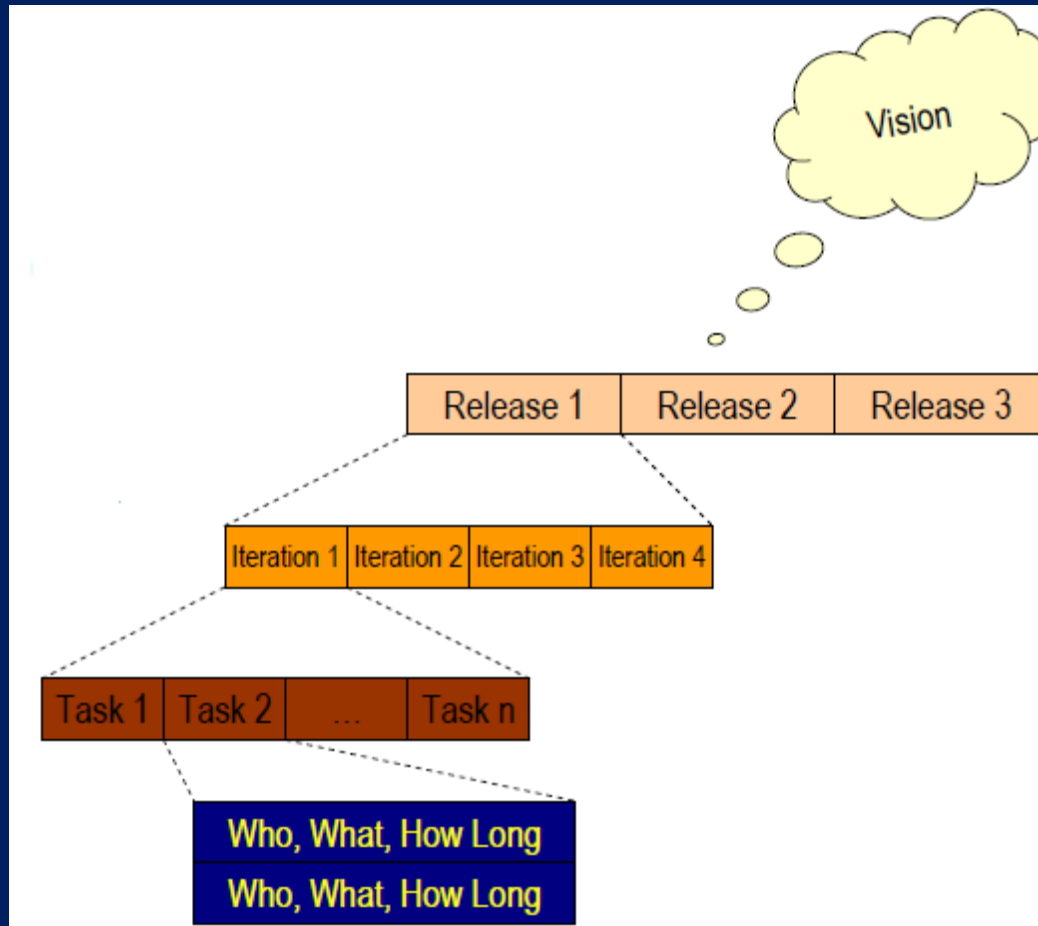
Iteration Plan



Iteration planning

- ▣ An **iteration planning** meeting is called at the beginning of each iteration to produce that iteration's plan of programming tasks
- ▣ Each iteration is 1 to 3 weeks long
- ▣ User stories are chosen for this iteration by the customer from the release plan in order of the most valuable to the customer first.
- ▣ Stories are further broken down into programming Tasks, each of 1 to 3 ideal days duration

XP Project



Iteration planning

- ▣ During Iteration Planning, the **Customer** presents the features desired for the next two weeks. The **programmers** break them down into tasks, and estimate their cost. Based on the amount of work accomplished in the previous iteration, **the team** signs up for what will be undertaken in the current iteration.

Iteration planning

- ▣ This is the first level where you actually commit (at the team level). You plan out what you're going to accomplish in the next iteration (a matter of weeks). You get a clear understanding on what done is (acceptance criteria).

Iteration plan

Iteration plan

A stack of stories and a stack of tasks.
Programmers sign up for tasks and estimate them

Iteration plan

- ▣ Unlike the release plan, the iteration plan is very much the **developers'**. They decide how to do things and in what order. The customer is still involved. It's important to report progress mid-iteration so that the customer can see what's happening and get a sense for what will actually happen.
- ▣ The **customer** will also get involved if the team finds there is too much to do and you need to cut scope.

Iteration plan

A *task* is the basic unit of work of an iteration. The stories are broken down into a list of tasks during the iteration planning meeting.

▣ For example, here are the tasks for of the stories from Release Planning:

▣ Close Chase

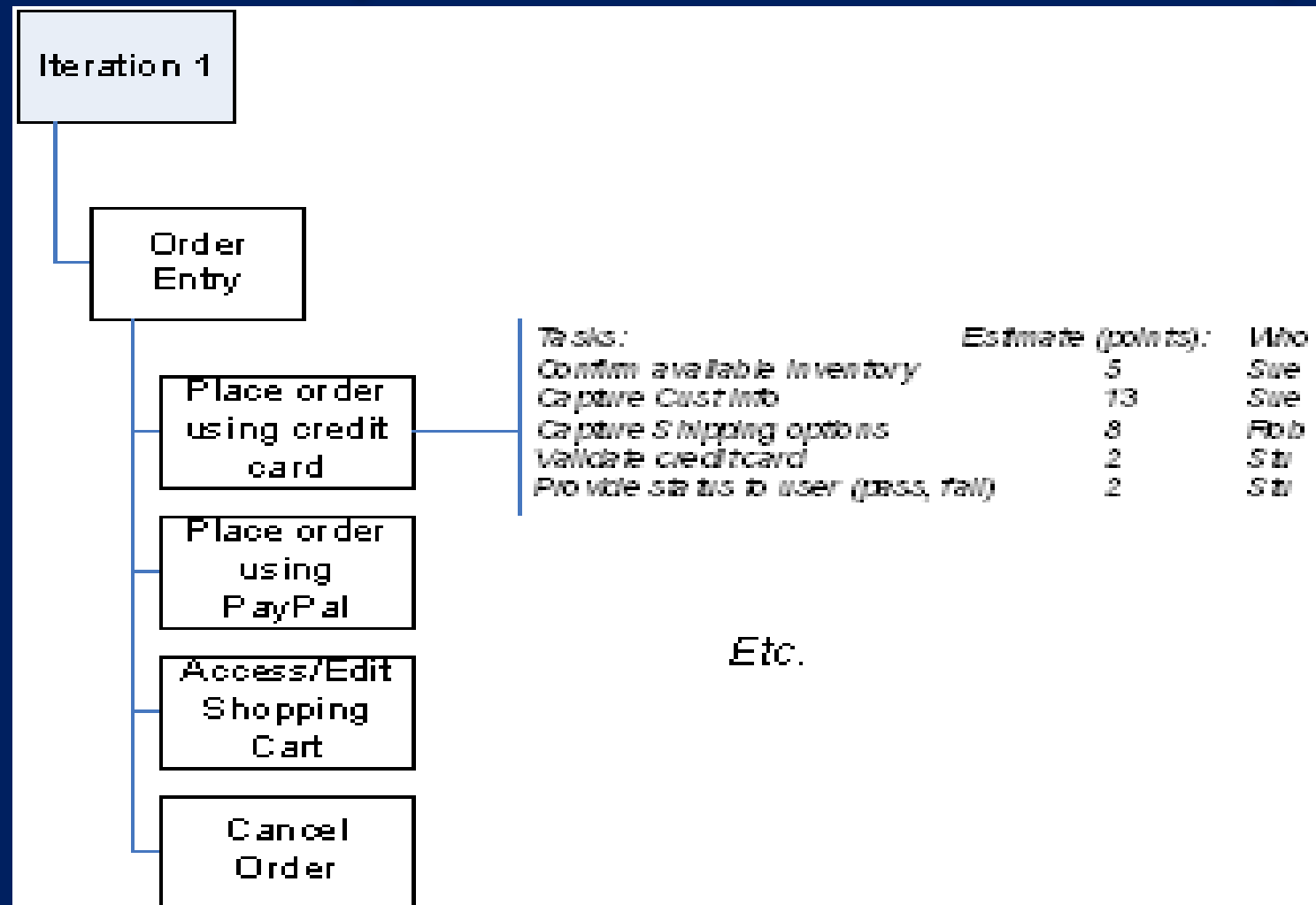
- Compare rates and services at local banks.
- Open new account.
- Move automatic payments (FedEx, EarthLink)
- After last check clears, transfer remaining funds.
- Acceptance test: verify account is closed and automatic payments working.

Iteration plan - Example

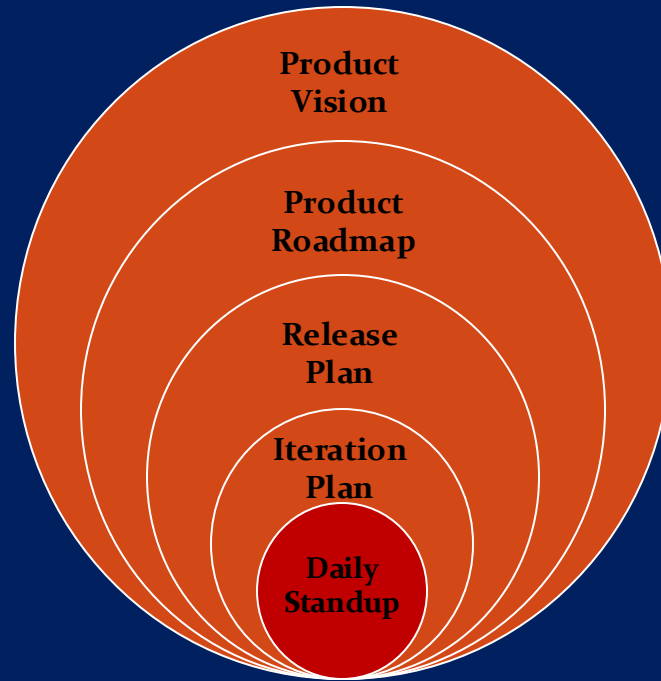
Find lowest fare (story)

Task	Estimated time (weeks)	Who
Alternative fare finder object	2	KB
Find candidate fares by date range	1	MF
Update ports to find alternatives	1	KB
Find candidate fares for alternative ports	1	KB
Special offers — major airlines lines	2	MF
Special offers — low-price airlines lines	3	RJ
UI (User Interface) for low fares	1	RJ

Iteration plan – Another example



Daily Standup



Daily Standup

- ▣ This is the level at which individuals commit to what they're going to get done in the next day.
- ▣ It helps to ensure that the team is communicating.
- ▣ It encourages collaboration and highlights issues which are blocking the team

Daily Standup

- ▣ Have a short meeting once a day so everybody knows what's going on, and what's not.
- ▣ The stand-up meeting is part of everyday life for agile teams. This daily meeting is not often seen as a planning session, but it certainly is.
- ▣ The people look a day ahead, have learned from the earlier days in the iteration, and tell each other what they plan on doing. Issues are raised, possibly addressed, and the success of delivering the desired features within the iteration can be determined after the meeting.



Daily Standup

- ▣ The format is simple. Everyone stands in a circle and you go round person by person.
- ▣ Each person briefly says what he or she did yesterday and what he or she is doing today.
- ▣ Problems or announcements that are important to the team are passed on.